NOETICA MASTER BLUEPRINT

PART 09

V2 Architecture - Scientific Intelligence Platform

Purpose

V2 transforms Noetica from a research intelligence delivery system into an interactive scientific intelligence platform.

V1 proves:

- People want scientific intelligence.
- Discovery ranking works.
- Personalization works.
- Reports provide value.

V2 introduces:

- Persistent user accounts
- Interactive discovery exploration
- Knowledge graph navigation
- Patent intelligence
- Grant intelligence
- Open-source intelligence
- Alert systems
- Public APIs

The goal is to move beyond email and build the first version of the Noetica platform.

Evolution

V1

Research Intelligence Reports

↓

V2

Scientific Intelligence Platform

↓

V3

Global Scientific Intelligence Network

Primary Objectives

1. Create an interactive platform.
2. Enable discovery exploration.
3. Implement knowledge graph navigation.
4. Introduce user accounts.
5. Track organizations and researchers.
6. Track patents and grants.
7. Launch the Noetica API.
8. Establish forecasting foundations.

Infrastructure Stack

Frontend

- Next.js
- React
- TypeScript

Backend

- FastAPI
- Python

Database

- PostgreSQL

Knowledge Graph

- PostgreSQL Relationships

- NetworkX

Email

- Resend

Authentication

- Auth.js
- Clerk
- Supabase Auth

Hosting

- Railway
- Render
- Fly.io

Source Control

- GitHub

Why V2 Exists

Email reports answer:

"What happened?"

V2 answers:

"What happened?"

"What does it connect to?"

"What influenced it?"

"What might happen next?"

Core Platform Modules

Discovery Explorer

Purpose:

Allow users to explore discoveries.

Users can search:

- Discoveries
- Fields
- Technologies
- Researchers
- Organizations

Example

Search:

CRISPR

Output:

Summary

Significance Scores

Related Discoveries

Patents

Grants

Organizations

Researchers

Knowledge Graph

Lifecycle Stage

Forecasts

Discovery Pages

Every discovery receives a dedicated page.

Discovery Page Components

Discovery Name

Description

Scientific Significance Score

Technological Significance Score

Civilizational Significance Score

Future Potential Score

Discovery Lifecycle

Related Discoveries

Knowledge Graph

Funding Activity

Patent Activity

Datasets

Research Timeline

Forecasts

User Accounts

Purpose:

Personalized scientific intelligence.

Stored Information

Email

Username

Expertise Level

Interests

Reading Time

Alert Preferences

Saved Discoveries

Feedback History

Watchlists

Personalization Engine V2

Inputs

Interests

Expertise

Activity

Feedback

Reading Patterns

Saved Discoveries

Outputs

Personalized Feed

Personalized Reports

Personalized Forecasts

Personalized Alerts

Forced Exploration Engine

Default Allocation

80%

User Interests

20%

Outside Interests

Purpose

Prevent intellectual echo chambers.

Promote cross-disciplinary discovery.

Discovery Watchlists

Users can follow:

Fields

Subfields

Discoveries

Researchers

Organizations

Technologies

Examples

CRISPR

Quantum Computing

AlphaFold

Synthetic Biology

Information Theory

DeepMind

CERN

Alert Engine

Purpose

Notify users about important events.

Alert Examples

Notify me when:

- New Quantum Computing breakthrough appears.
- New Bioinformatics discovery appears.
- New field emerges.
- New patent cluster forms.
- Funding momentum exceeds threshold.
- Discovery significance score changes substantially.

Knowledge Graph Explorer

Purpose

Enable navigation of connected knowledge.

Example

Information Theory

↓

Machine Learning

↓

Transformers

↓

Protein Language Models

↓

Drug Discovery

Users can explore:

Parent Nodes

Child Nodes

Influencing Discoveries

Enabled Discoveries

Competing Discoveries

Adjacent Fields

Knowledge Graph Storage

V2 Implementation

PostgreSQL

+

NetworkX

Stores

Nodes

Relationships

Weights

Influence Scores

Connectivity Scores

Researcher Intelligence Layer

Track:

Researchers

Inventors

Scientists

Research Groups

Laboratories

Researcher Metrics

Discovery Contributions

Cross-Field Influence

Research Momentum

Emerging Influence

Collaboration Networks

Organization Intelligence Layer

Track

Universities

Research Institutes

Companies

Government Agencies

Think Tanks

Nonprofits

Examples

MIT

Stanford

CERN

NIH

EMBL

DeepMind

OpenAI

Anthropic

Organization Metrics

Discovery Output

Funding

Patent Activity

Research Growth

Cross-Field Impact

Influence

Patent Intelligence Layer

Purpose

Track technological significance.

Sources

USPTO

WIPO

Google Patents

European Patent Office

Patent Metrics

Patent Growth

Patent Clusters

Commercialization Signals

Technology Diffusion

Emerging Technology Indicators

Grant Intelligence Layer

Purpose

Track future scientific direction.

Sources

NIH

NSF

ERC

Horizon Europe

Government Programs

National Research Missions

Metrics

Funding Growth

Research Momentum

Emerging Fields

Institutional Support

Strategic Importance

Conference Intelligence Layer

Purpose

Detect breakthroughs before publication.

Sources

NeurIPS

ICML

ICLR

ISMB

RECOMB

APS

ACS

Nature Conferences

AAAI

CVPR

ECCV

EMNLP

Metrics

Emerging Topics

Workshop Growth

Keynote Themes

Research Momentum

Novelty Signals

Open Source Intelligence Layer

Purpose

Track practical adoption.

Sources

GitHub

GitLab

Hugging Face

Open Research Repositories

Metrics

Stars

Forks

Contributors

Issue Activity

Adoption Growth

Community Expansion

Dataset Intelligence Layer

Purpose

Track foundational research assets.

Examples

Human Genome

TCGA

AlphaFold DB

ImageNet

Protein Data Bank

Large Language Model Datasets

Metrics

Usage

Citations

Derivative Discoveries

Field Influence

Research Dependence

Discovery Momentum Engine

Measures

Publication Growth

Citation Growth

Funding Growth

Patent Growth

Open Source Growth

Research Community Growth

Output

Momentum Score

Emerging Field Detector

Questions

What fields are appearing?

What fields are merging?

What fields are declining?

What fields are accelerating?

Examples

AI Drug Discovery

Protein Language Models

Quantum Biology

Autonomous Science

Scientific Foundation Models

Forecasting Engine V1

Purpose

Estimate future importance.

Questions

Could this become foundational?

Could this create a new field?

Could this transform multiple disciplines?

Could this generate new technologies?

Outputs

Future Potential Score

Probability Estimates

Confidence Intervals

Alternative Scenarios

Public API

Purpose

Allow developers to build on Noetica.

API Categories

Discoveries

Fields

Researchers

Organizations

Patents

Grants

Forecasts

Knowledge Graph

Statistics

Reports

Weekly Intelligence Summary

Contents

Top Discoveries

Top Emerging Fields

Important Patents

Important Grants

Open Source Signals

Forecast Updates

Research Momentum Changes

Monthly Frontier Report

Contents

Most Significant Discoveries

Fastest Growing Fields

Major Cross-Disciplinary Breakthroughs

Technology Convergence Signals

Scientific Forecast Updates

Annual State of Human Knowledge

Contents

Most Important Discoveries

Most Important Emerging Fields

Most Influential Researchers

Most Influential Organizations

Most Influential Datasets

Most Influential Technologies

Most Important Forecast Signals

Predictions For Next Year

V2 Success Criteria

Users can:

Search discoveries.

Explore knowledge graphs.

Track researchers.

Track organizations.

Follow fields.

Create alerts.

Explore patents.

Explore grants.

Access forecasts.

Receive personalized intelligence.

End State

V2 transforms Noetica from a research reporting system into an interactive scientific intelligence platform capable of tracking, connecting, and explaining discoveries across the entire human knowledge ecosystem.

The platform becomes the first operational version of Noetica's long-term vision: a living map of human knowledge.